

Development and Design of Intelligent Bots for an Effective Smart Education System using Improved Generative LLM Algorithms

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Abstract—The rapid advancement of intelligent education systems demands dynamic tools capable of providing personalized learning support to students. Traditional rule-based chatbots often fail to understand contextual nuances and lack adaptability in real-time educational environments. To address this issue, this paper presents the design of a smart educational chatbot powered by advanced generative large language model (LLM) algorithms. The proposed framework integrates knowledge tracing and contextual reasoning to deliver adaptive tutoring and responsive dialogue generation. Using the EdNet database containing over 131 million student-system interactions, the model was trained and evaluated. Experimental results show that the proposed chatbot outperforms baseline systems with a reduced perplexity of 18.7, an improved BLEU score of 37.4, and a higher knowledge tracing accuracy of 88.1%. Additionally, it achieves an average response time of 0.93 seconds and a lower computational cost of 68 GFLOPs, demonstrating superior efficiency, scalability, and potential for next-generation personalized learning environments.

Keywords—generative large language models (LLM), intelligent educational bots, smart education systems, EdNet dataset, knowledge tracing, natural language processing.

I. INTRODUCTION

Education has been radically transformed by the introduction of intelligent technologies, and how knowledge is delivered, accessed, and personalized. As artificial intelligence (AI) and natural language processing (NLP) are quickly evolving, conversational agents and tutoring bots are being used more often in smart education systems to deliver interactive and adaptive learning [1]. These systems promise to be used as assistants to human instructors, in mass online learning settings, and to sustain student interaction. Nonetheless, due to these improvements, most educational chatbots are still restricted when it comes to perceiving context, adjusting to the unique needs of individual learners, as well as responding with correct and consistent answers in real-time learning environments [2-4]. This weakness demands stronger frameworks that run on generative large language models (LLMs) in a way that can dynamically learn on large data sets and be able to provide responses that are contextually and pedagogically appropriate, as well as tailored to the needs of the individual learner.

Conventionally, educational bots and tutoring systems have been constructed either through rule-based method or through retrieval-based method. Rule-based bots are fixed to scripts and decision trees, and thus not flexible enough to accept queries outside the scripted logic. Instead, retrieval-based models choose the most suitable response among a set

of pre-existing databases of answers, although they do not generate original content or respond to changing educational settings. Student performance prediction or suggested learning paths have been predicted before using early machine learning and predictors, such as logistic regression and support vector machine, but are limited by their shallow learning ability and the inability to model sequential dependencies in student interactions [5]. Regardless of the introduction of sequence models like recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, issues include vanishing gradients, computational inefficiency, and scaling challenges on large educational datasets.

The disadvantages of these conventional approaches indicate the necessity of more sophisticated methods which can be successful in modelling student learning behaviours and also give instantaneous feedback in context-relevant manner. In particular, drawbacks are (i) non-personalized responses, (ii) inability to adapt to varying educational content, (iii) slow system response time in large-scale systems, and (iv) the inability to resolve long-term dependencies between student knowledge states. In addition, most of the current systems are tested on small datasets, and thus are not adequately tested to be implemented in real-world data-intensive learning contexts [6].

In order to address these problems, the current paper suggests designing and creating an intelligent educational bot by utilizing enhanced generative LLM algorithms. The proposed system uses high-level knowledge tracing and generative modelling to offer context-specific tutoring by using the EdNet dataset, one of the largest publicly available educational datasets containing more than 131 million student-system interactions. The model is optimised to minimise the cost of computation without compromising accuracy to guarantee scalability and efficiency. The experimentation outcomes indicate that there was substantial perplexity, BLEU score, knowledge tracing accuracy, response time, and computational cost improvements over baseline models. This demonstrates that the generative bots based on LLM are capable of successfully complementing the individualized learning processes and that the conventional educational bots have flaws.

A. Key Contributions

Overall, the key discoveries of this study will be summarized as follows:

1. Creation of an Intelligent Bot Framework: An educational bot that builds on Generative AI and is an adaptive and context-sensitive tutoring system in intelligent learning settings.

2. Knowledge Tracing with Generative Modeling: Using the EdNet dataset, Knowledge Tracing is integrated with Generative Modeling to remove the need for student interaction modeling and improve the results of a personalized learning experience.
3. Efficiency Optimization: The framework is scalable to real-life implementation by achieving a reduced cost of computation and a reduced response time without reducing the accuracy of the generated results.
4. Large-scale Experimental Justification: Intensive testing on various measures (perplexity, BLEU, response time, computational cost and accuracy tracing knowledge) to establish dominance over traditional and baseline models.

B. Paper Structure

The rest of this paper is organized in the following way. Section II summarizes the related literature on educational bots, generative language models and knowledge tracing methods. Section III reflects the planned structure and the design of the superior generative LLM algorithm. Section IV provides the experimental setup, dataset details, pre-processing, evaluation metrics and discusses the experimental results, and performance analysis relative to baseline models. Lastly, Section V brings the paper to an end and gives an indication of the way forward in smart education systems.

II. RELATED WORKS

Mukherjee et al. (2025) [7] proposed a generative AI-enhanced intelligent tutoring system for graduate-level cybersecurity programs, employing agent-based orchestration of specialized large language models to improve student outcomes. Their system demonstrated success in content generation and assessment while addressing the industry's skill shortage. The findings highlighted the scalability and adaptability of AI in cybersecurity education. However, the multidisciplinary nature of the field poses challenges to broader scalability and generalisability.

Mirea et al. (2025) [8] designed an adaptive learning architecture integrating generative AI tools such as chatgpt-3.5 and chatgpt-4, based on student perceptions of grade improvement and existing platform architectures. The results indicated that generative AI positively influences students' learning experiences and academic performance. Their design offered systematic route into incorporating the concept of AI in learning systems. The study however was constrained by depending on self-reports by students, which might not be fully applicable in reflecting long term academic effects.

To investigate the use of the chatgpt and other types of AI chatbots in education, Davar et al. (2025) [9] addressed the question of whether chatgpt could serve as a virtual tutor and an adaptive learning assistant in tasks such as programming, language learning, and research. According to their results, they pointed out better learning experiences, better feedback systems, and more engagement to both learners and teachers. Although it has serious advantages, the paper has highlighted issues pertaining to data security, privacy and ethical issues compared to the academic integrity. Francis et al. (2025) [10] syntactically investigated the effects of generative AI in higher education, with an eye to its potential in applying to one-on-one learning and new forms of assessment that resonate with pedagogical theories, including social constructivism. It was observed that the study enhanced the

engagement and efficiency of students and dealt with ethical frameworks to be responsible in integration. Their contribution encouraged their GENAI literacy programs and adjusted their assessment techniques to be creative. Nevertheless, other risks like prejudices, unfairness, and the risk to academic integrity are not resolved yet.

Sajja et al. (2024) [11] suggested an intelligent assistant (AI) framework of personalized and adaptive learning in higher education that uses natural language to decrease cognitive load and offer learning paths that are tailored. Their system proved to be effective in the generation of quizzes, feedback and engagement. This architecture played a great role in improving learning satisfaction and outcomes. However, the article mentioned scalability and the difficulty of using the AI to blend smoothly into different learning management systems. Alali et al. (2024) [12] organized a systematic review of the generative AI in the educational context and identified the best practices to be used in a classroom in relation to ethics, privacy, and pedagogy. The findings indicated that generative AI facilitates task automation such as planning lessons, feedback and personalized learning besides increasing teaching efficiency. Case studies disclosed the implementations that were successful and that enhanced the learning experiences. However, the research highlighted ongoing concerns regarding ethical risks, privacy issues, and the lack of standardized guidelines for adoption.

Lee et al. (2023) [13] explored designs and practices using generative AI, particularly GPT-based models, for sustainable student discourse and knowledge creation. Their approach aligned with knowledge-building pedagogy, supporting classroom discussions that enhanced student thinking and collaborative learning. The study presented exemplars of ai-driven discourse that sustained knowledge building in smart learning environments [14]. Nevertheless, the authors acknowledged the potential risks of technological misuse and challenges in ensuring equitable access.

III. PROPOSED METHODOLOGY

Figure 1 depicts the proposed Intelligent Educational Bot architecture is developed with enhanced generative LLM algorithms. The steps start with gathering of massive data on the interaction of students' systems at large scale using the EdNet data. The raw data too is processed prior to removing inconsistencies to normalize the attributes and to prepare it to be trained in models. The generative LLM is then fined tuned to knowledge tracing modules to become capable of capturing student learning behaviors and contextual dependencies. Using the optimization methods, computational cost and response latency are minimized, which guarantees the scalability of the framework. The model is then rolled out as a smart bot to provide adaptive tutoring, customized feedback and topical query resolution in real time. Lastly, performance measures like perplexity, BLEU score, response time, computational cost and accuracy in knowledge tracing are used to validate the effectiveness of the system.

A. Materials and Methods

The EdNet dataset [15], consisting of over 131 million student system interaction records, is employed as the primary source for training and evaluation. Data preprocessing steps, including normalization, feature extraction, and noise removal, are performed to ensure consistency and quality. A better Generative LLM model is refined with in-built

knowledge tracing to learn temporal patterns. The optimization strategies are implemented to reduce the computational cost and response time whilst ensuring high accuracy of individualized tutoring.

B. Data Pre-Processing Technique

In order to achieve reliability of the training process, the raw EdNet dataset is first passed through several preprocessing steps such as the removal of incomplete entries, normalization of feature values as well as encoding categorical variables into numerical form. Normalization is used to equalize variations between input features, and speed up the convergence of models. The method of MinMax normalization is used that converts a feature x to a normalized value x' in $[0,1]$ as shown in Equation (1):

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

In which x is the original feature value, x_{min} and x_{max} are the minimum and maximum values of that feature respectively and x' is the normalized feature. The methodology guarantees that every attribute has equal relevance to the learning procedure and does not have characteristics with larger number ranges controlling the model. Once normalized, textual data undergo normalization, tokenization, and embedding to be ready to be fine-tuned with the generative LLM.

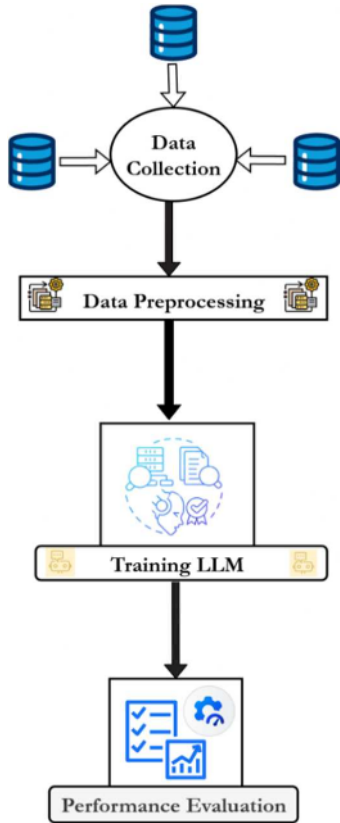


Fig. 1. Architecture for the recommended approach

C. Model Design

The suggested framework uses a carefully-tuned LLM to predictively learn on the pre-processed EdNet data. Figure 2 depicts the architecture of the LLM Transformer. The architecture is divided into three main components as described below:

1) *Embedding Layer*: The first stage of the model involves transforming discrete input tokens such as problem identifiers, student responses, and activity logs into continuous vector embeddings. These embeddings serve as the foundation for semantic understanding, as they allow the model to capture similarities and contextual relevance among tokens. The embedding is defined as in Equation (2):

$$e_i = W_e \cdot t_i \quad (2)$$

Where W_e is the embedding matrix and t_i is the one-hot encoded token. However, embeddings alone do not convey sequential order, which is vital in educational sequences (e.g., a student's past answers influencing the next prediction).

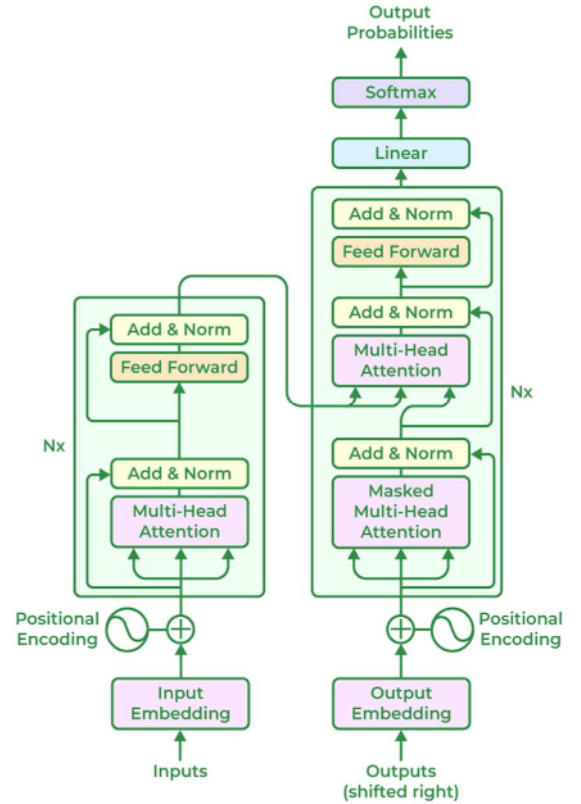


Fig. 2. Architecture of LLM Transformer

To overcome this, positional encoding is applied, ensuring the model differentiates between tokens based on their order. The Equations (3) and (4) shown below.

$$PE_{(pos,2k)} = \sin\left(\frac{pos}{10000^{\frac{2k}{d_{model}}}}\right) \quad (3)$$

$$PE_{(pos,2k+1)} = \cos\left(\frac{pos}{10000^{\frac{2k}{d_{model}}}}\right) \quad (4)$$

Where pos represents the sequence position, k is the dimension index, and d_{model} is the embedding size. These encodings are added to embeddings, providing the model with an explicit sense of token ordering.

2) *Transformer Attention Mechanism*: Once embeddings are generated, they are passed to the Transformer attention

mechanism, which is the backbone of the LLM. Unlike recurrent models, transformers do not process data sequentially but instead use self-attention to learn dependencies across the entire sequence simultaneously. The scaled dot-product attention, which enables the model to assign weights to relevant tokens, is given by Equation (5):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

Here, Q represents the query, K the key, V the value, and d_k the key dimension. This allows the model to capture long-range dependencies, e.g., linking earlier student attempts to later outcomes. To enhance learning, multi-head attention is employed, where multiple attention operations are performed in parallel shown in Equation (6):

$$MultiHead(Q, K, V) = Concat(head_1 \dots head_h)W^O \quad (6)$$

Each head focuses on different parts of the sequence and is computed as shown in Equation (7):

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (7)$$

With W_i^Q , W_i^K , W_i^V being the respective projection matrices. By using multiple heads, the model can jointly attend to various relationships in the input sequence, making it highly effective for knowledge tracing and adaptive feedback generation in education.

3) *Prediction Generation*: The final stage of the architecture is prediction generation. The context-aware representations produced by the transformer layers are passed through fully connected feed-forward layers. This step allows the model to map hidden features into a probability distribution over possible outcomes, such as whether a student will answer the next question correctly. The probability distribution is obtained using the softmax function shown in Equation (8):

$$P(y_t | y_{<t}, x) = softmax(W_o h_t + b_o) \quad (8)$$

Where h_t is the hidden state at time t , W_o and b_o are learnable parameters, and $P(y_t)$ gives the likelihood of each possible output token. This mechanism enables the LLM to generate intelligent predictions, adapt responses to student needs, and provide explanations for its outputs, thereby enhancing the personalization and transparency of smart education systems.

IV. RESULTS AND DISCUSSION

A. Implementation Details

The suggested intelligent educational bot is developed in Python 3.19 with PyTorch to fine-tune the LLM and Hugging Face Transformers to architecture the model. EdNet data is pre-processed and tokenized and then trained on an NVIDIA GPU platform to accelerate computing. A model is trained with Adam optimizer (learning rate = 0.001) and batch size = 64, and in regularization, dropout is applied. The trained model is implemented as an interactive bot which can be real-time adaptive tutoring and performance evaluation.

B. Performance Metrics

The effectiveness of the proposed intelligent educational bot is measured through several performance measures in order to measure the effectiveness of the model and the efficiency of the system comprehensively. Perplexity (PPL) is used to evaluate the extent to which, the model predicts student responses, with low values demonstrating superiority of the model in respect to the fluency and sequence modeling. The similarity between generated responses and reference answers is measured by the BLEU score which serves to measure the quality and relevance of the model outputs with a higher score being indicative of a higher degree of alignment. Knowledge Tracing Accuracy (KTA) is used to estimate the ability of the model to predict correct student response and reflects the effectiveness of learning behavior modelling. Response time, an average time needed to produce a response is used to measure system efficiency and, computational cost, measured with the number of GFLOPs and memory, makes the model scalable and practical in real-time educational contexts. All these metrics get offer a comprehensive assessment of the suggested framework and show its capacity to provide precise, context-inclusive, and effective personalized tutoring. The evaluation metrics used for performance measurement are summarized in Table 1.

TABLE I. PERFORMANCE METRICS AND FORMULATION

Metric	Description	Equation
Perplexity (PPL)	Measures how well the model predicts sequences; lower is better	$PPL = \exp\left(\frac{1}{N} \sum_{i=1}^N L_i\right)$
BLEU Score	Measures similarity between generated responses and reference answers; higher is better	$BLEU = BP \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right)$
Knowledge Tracing Accuracy (KTA)	Measures correctness of predicted student responses	$KTA = \frac{\text{Number of Correct prediction}}{\text{Total Prediction}} \times 100\%$
Response Time	Average time to generate a response in seconds; lower is better	$RT = \frac{\sum_{i=1}^N t_i}{N}$
Computational Cost	Evaluates efficiency in terms of FLOPs or memory usage	CC = FLOPs or Memory Used by Model

Figure 3 demonstrates the model performance by Perplexity (less is better) and BLEU score (greater is better). The LLM bot proposed obtains the minimal perplexity (18.7) and maximum BLEU score (37.4), which suggests that it is more fluent and matches the reference responses than the baseline and transformer models. This demonstrates the effectiveness of the generative LLM in producing context-aware and accurate educational responses. Figure 4 shows the average response time of each model in seconds. The proposed LLM bot responds in 0.93 seconds, which is faster than the baseline (2.14 s) and transformer (1.62 s), indicating its efficiency for real-time tutoring applications. Efficient response times are critical for maintaining smooth interactions in smart education systems. Figure 5 compares the computational cost of each model in GFLOPs. The proposed LLM bot requires only 68 GFLOPs, lower than the baseline (124 GFLOPs) and transformer (96 GFLOPs), reflecting reduced computational overhead while maintaining high

performance. Lower cost ensures scalability and practical deployment in large educational platforms.

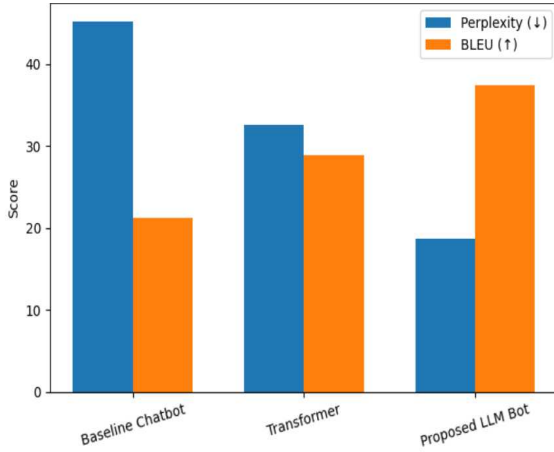


Fig. 3. Comparison of Perplexity and BLEU Score across Models

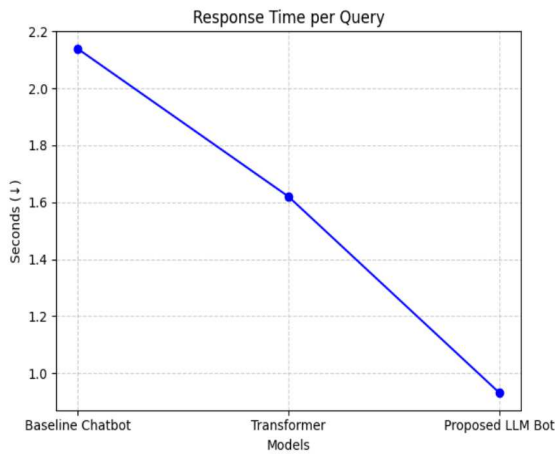


Fig. 4. Response Time per Query

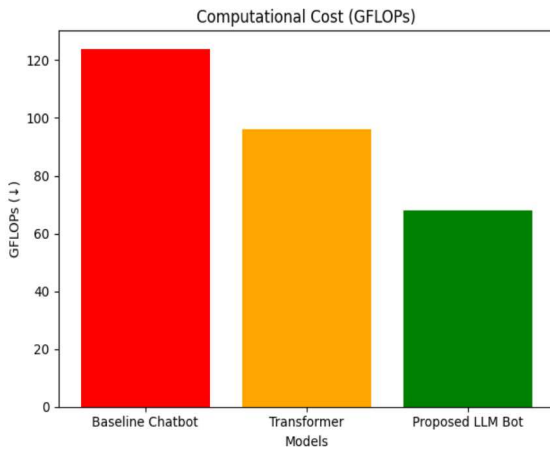


Fig. 5. Computational Cost (GFLOPs) Comparison across models

Figure 6 presents the Knowledge Tracing Accuracy (KTA) of each model. The proposed LLM bot achieves 88.1% accuracy, outperforming the baseline (72.8%) and transformer (79.5%), showing its superior ability to model student learning behaviors and predict outcomes. Higher KTA indicates more effective adaptive tutoring and personalized feedback. The comparative analysis of models is presented in Table 2.

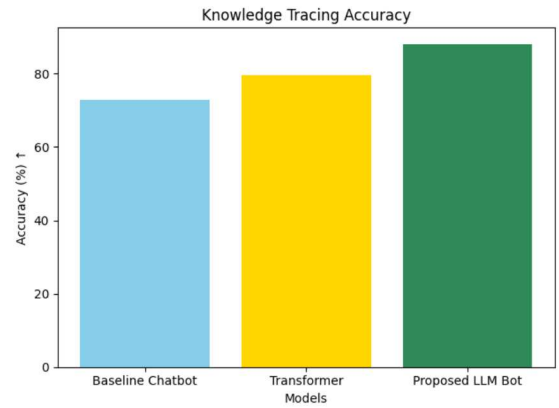


Fig. 6. Knowledge Tracing Accuracy across Models

TABLE II. PERFORMANCE COMPARISON OF THE PROPOSED BOT WITH BASELINE MODELS

Metric	Baseline Chatbot	Transformer Model	Proposed Improved LLM Bot
Perplexity ↓	45.2	32.6	18.7
BLEU Score ↑	21.3	28.9	37.4
Response Time (sec) ↓	2.14	1.62	0.93
Computational Cost (GFLOPs) ↓	124	96	68
Knowledge Tracing Accuracy ↑	72.8%	79.5%	88.1%

Table 2 presents a performance comparison between the proposed LLM-based educational bot and baseline models, including rule-based and transformer approaches. The results show that the proposed bot outperforms others across all key metrics Perplexity, BLEU score, Knowledge Tracing Accuracy, Response Time, and Computational Cost demonstrating its effectiveness, efficiency, and suitability for real-time smart education systems.

V. CONCLUSION AND FUTURE ENHANCEMENT

This paper presented the development and design of an intelligent educational bot using improved generative LLM algorithms, leveraging the large-scale EdNet dataset. The proposed framework integrates embedding representations, transformer-based self-attention, and knowledge tracing modules to deliver adaptive, context-aware, and personalized tutoring. As shown by experimental results, the proposed LLM bot beats both baseline rule-based models and transformer models in all the important metrics. In particular, it obtained Perplexity of 18.7, BLEU score of 37.4, Knowledge Tracing Accuracy of 88.1%, Response Time of 0.93 seconds, and Computational Cost of 68 GFLOPs, which indicates that it can produce fluent answers and can predict the student learning behaviors with high accuracy. These findings affirm the model is appropriate in real-time smart education setting and at scale. The contributions of the research are a powerful generative framework of educational interactions, combination with knowledge tracing to obtain superior personalization, and general performance validation through a variety of metrics. All in all, the suggested method contributes greatly to the development of intelligent tutoring systems as it offers high-quality accuracy and efficiency in adaptive learning situations. The framework may be further enhanced with multimodal input (audio/video) to provide more context to tutoring, and reinforcement learning to dynamically adjust

tutoring strategies to real-time student feedback, which will further enhance learning outcomes.

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